Truck-Fleet Risk Analysis

Agenda

- Problem Statement
- Process Flow
- Risk Model by Count
- Risk Model by Drivers
- Risk Model due to Incidents
- **Risk Model due to Speed**
- Conclusion

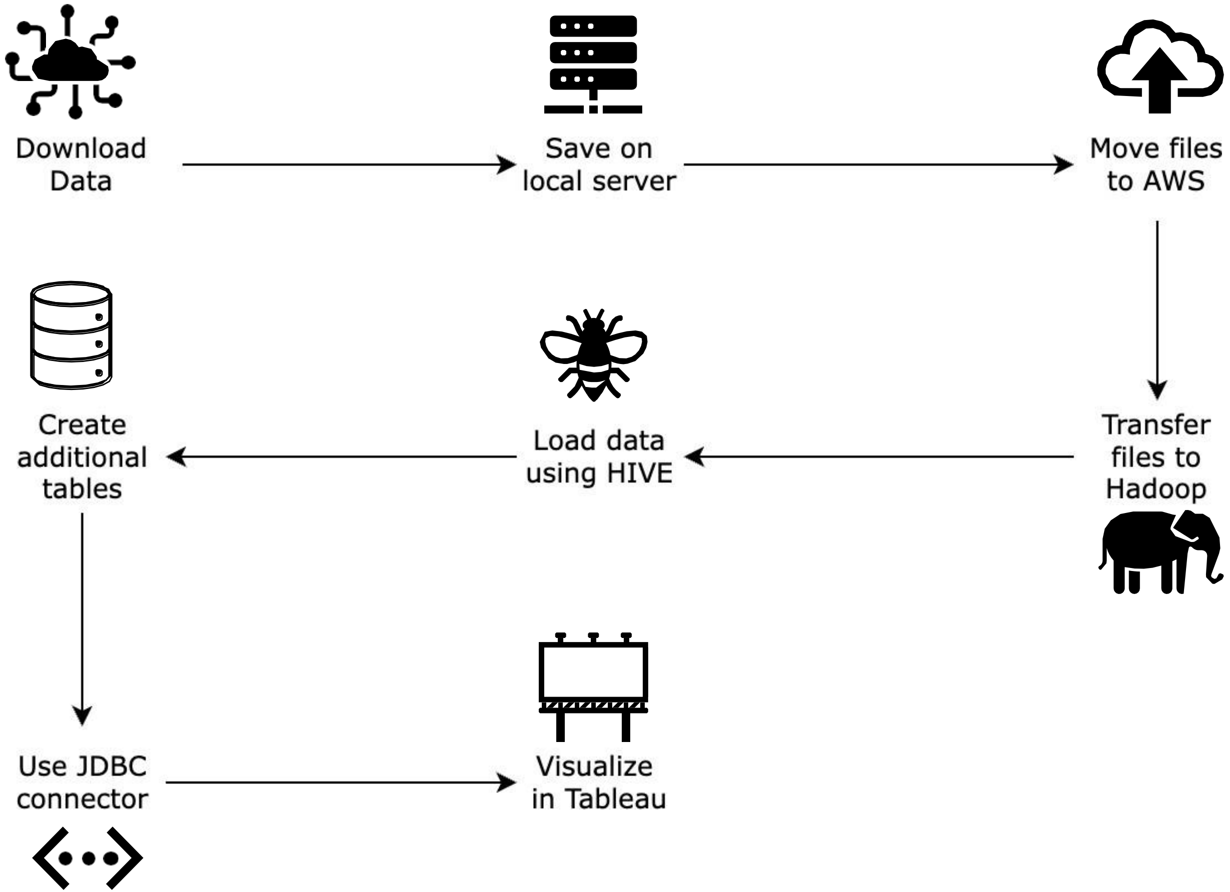
Problem Statement

To predict risk factors for drivers based on events, total distance, average speed, and mileage

Identify risky drivers, on a scale of 1 – 10, by analyzing data on location, vehicles, mileage, gas consumption, events, and risk factors

To reduce driver risks and help prevent accidents involving large commercial trucks in the US, minimizing injuries and deaths

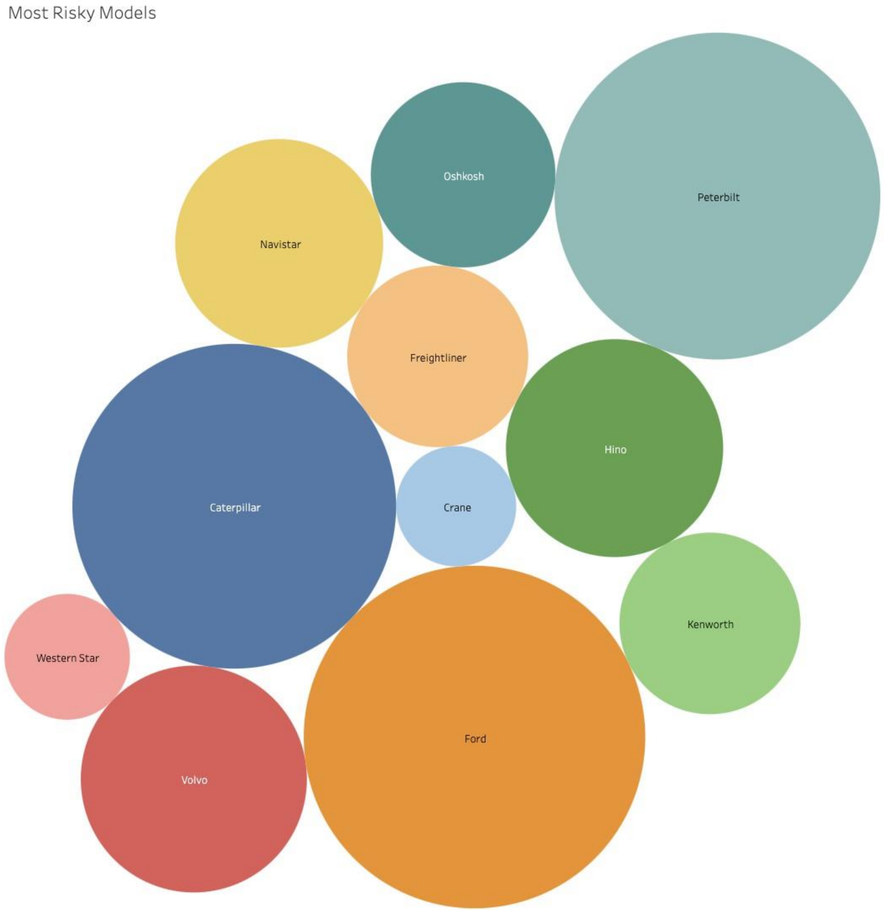
Process Flow



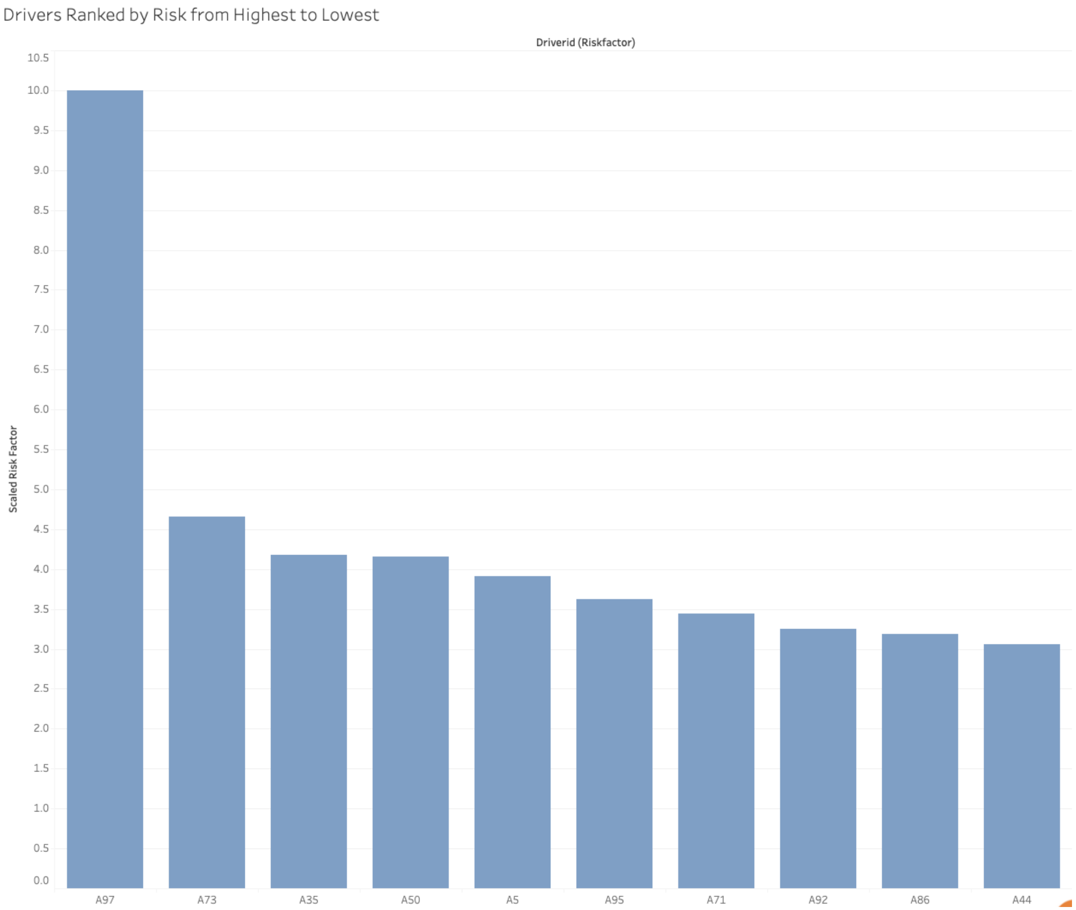
Dashboard 1 - **Risk Model by Count**

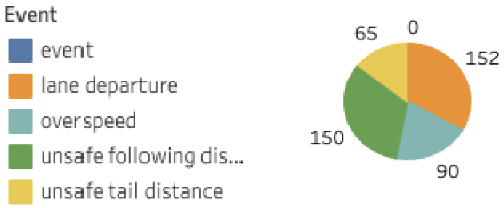
The color and size represents the event count by model

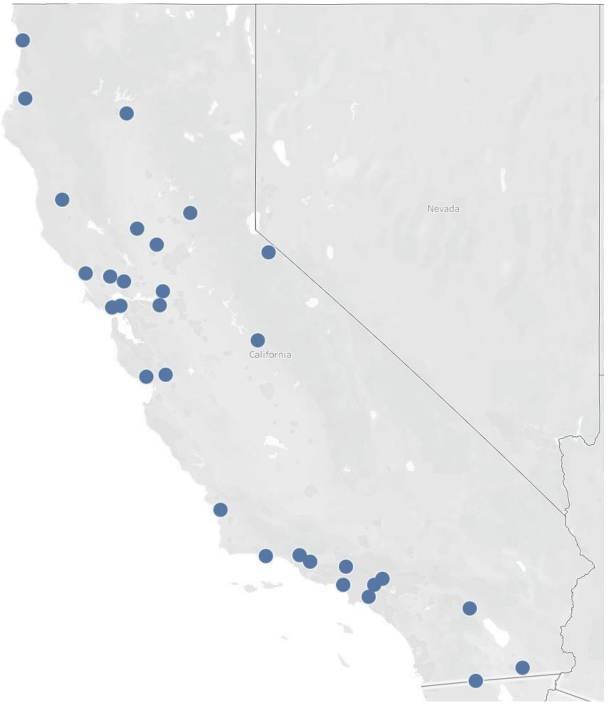
By observation – Ford, Caterpillar and Peterbilt are the largest representation



Dashboard 2 - **Risk Model by Drivers**







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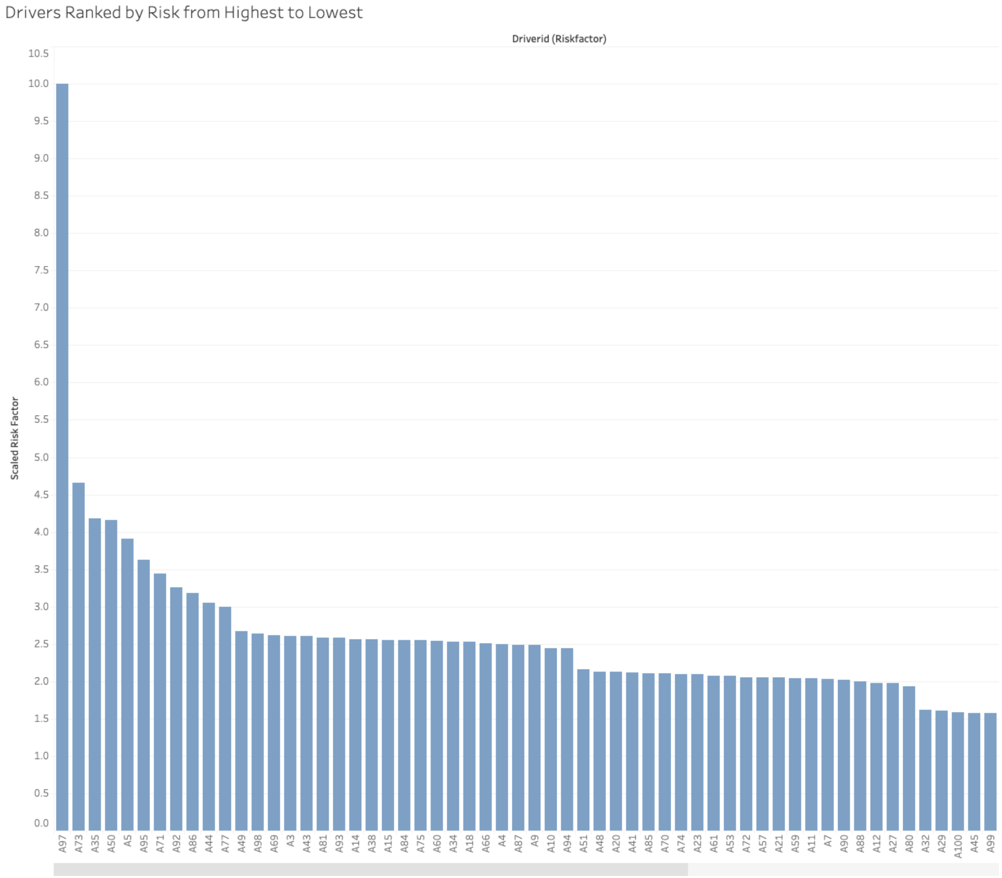
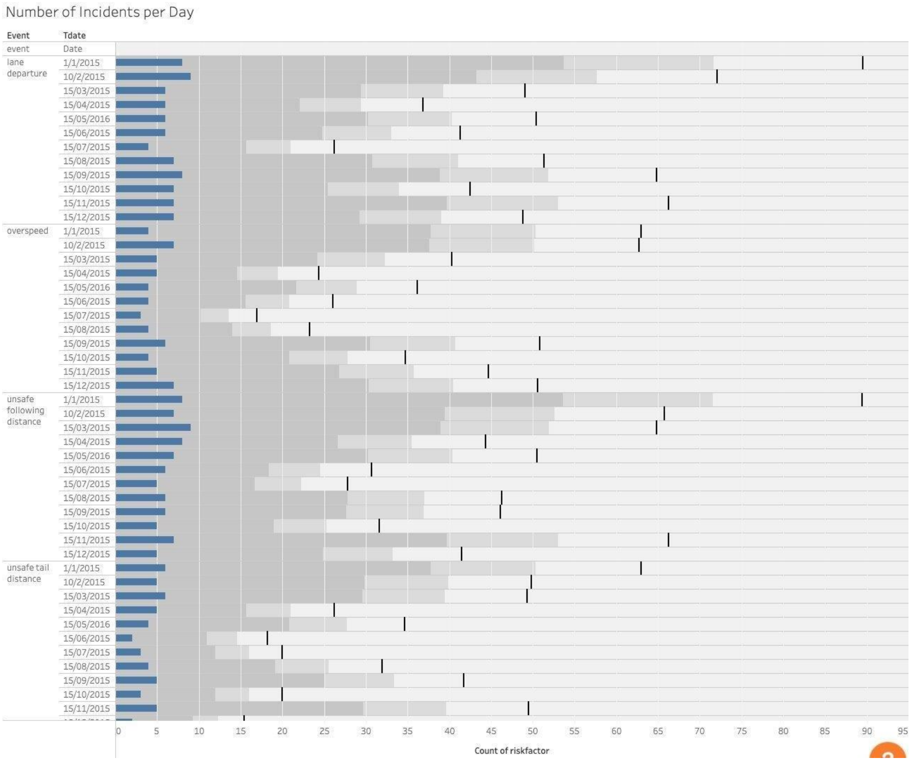


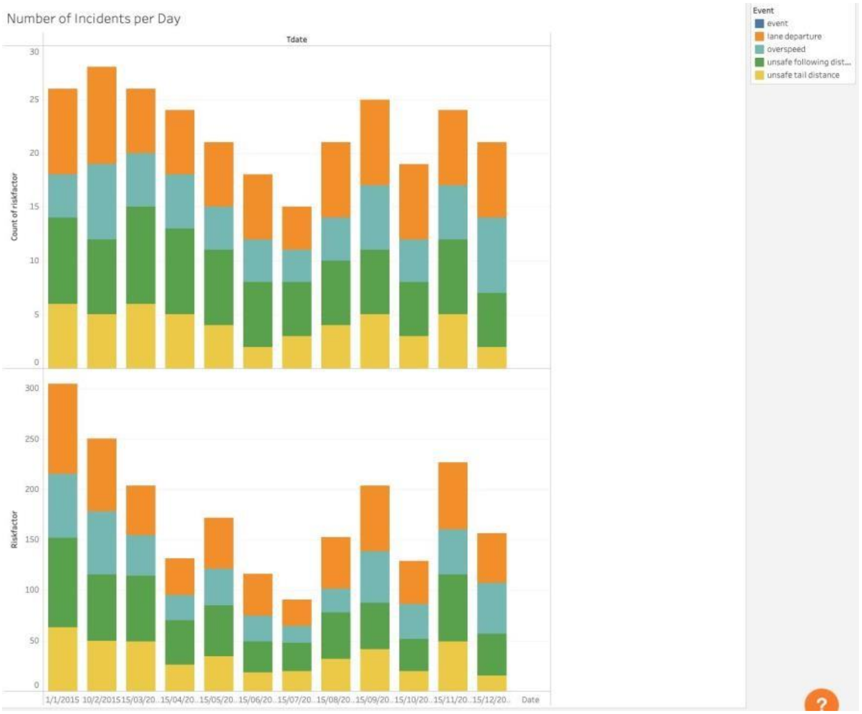
Model shows the scaled Risk Factor Analysis of drivers

The model highlights the riskiest driver, "A97", with a risk factor of 10, significantly higher than any other driver

The second-highest risk, associated with Driver "A73", is 4.6, indicating that Driver "A97" is more than twice as risky compared to "A73

Dashboard 3 - **Risk Model due to Incidents per Day**

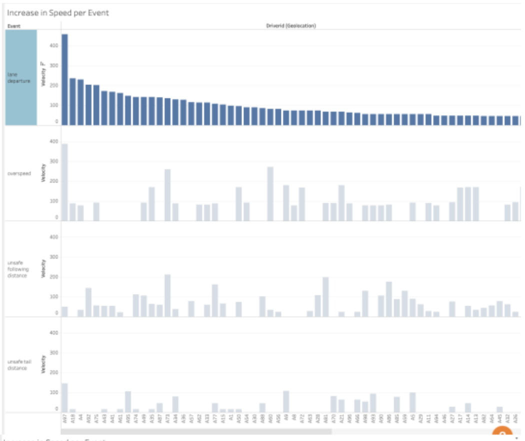


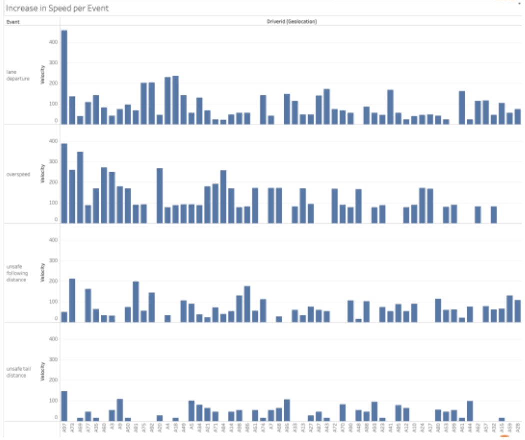


Dashboard 3 - **Risk Model of Top 10 Driver Events**

The model highlights that distribution of events per driver

Dashboard 4 - Risk Model due to Speed





Model shows the increase in speed per event

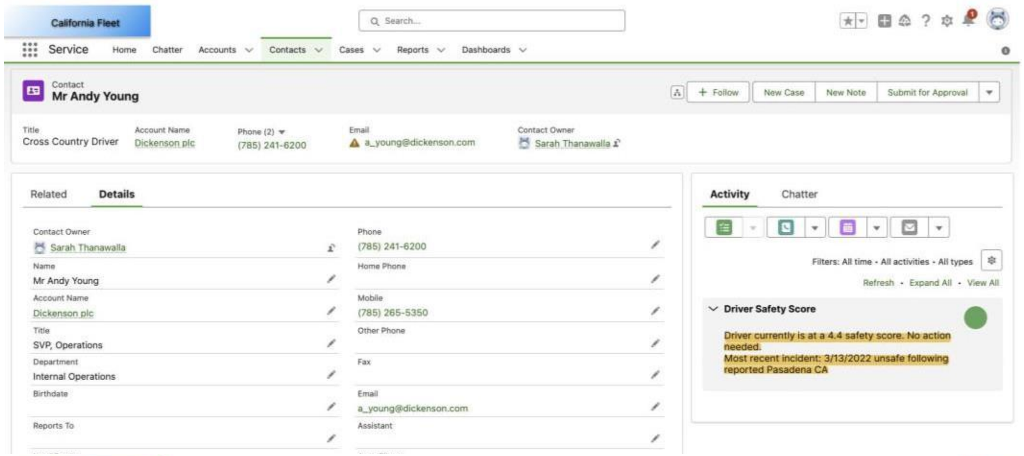
The model highlights a direct correlation between an increase of speed and the occurrence of an event

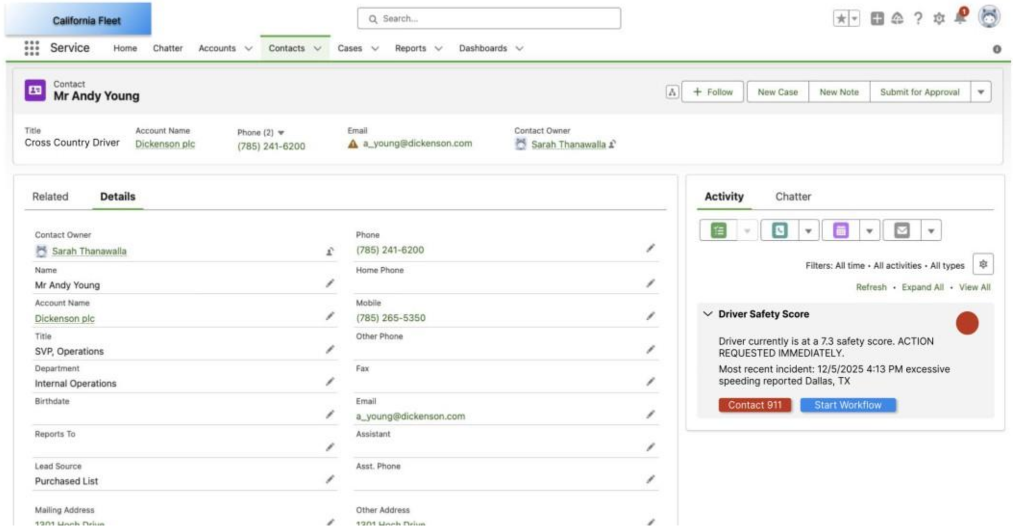
Conclusion

To mitigate risk, we should:

- Train drivers with the highest risk factors
- Raise awareness of safe driving practices
- Top rated drivers with the most miles to train high-risk drivers

Extra Analysis: Salesforce





Thank you